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Exploring Multi-Temporal Scale Co-Location of Childhood Respiratory Disease Incidents in Nanning City: A Guide to Geographically and Temporally Weighted Colocation Quotients

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Method Article

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Abstract

Background

The incidence of disease data occurring in close spatial and temporal proximity are likely to exhibit unobserved effects. Investigating the spatial and temporal associations among various categories of childhood respiratory diseases is a crucial for modelling of demographic, environmental, and behavioral factors influencing these diseases. Traditional spatial statistical methods that do not account for associations among incident categories risk producing spurious findings.

Methods

This paper presents a practical approach for effectively handling spatio-temporal incident disease data, with a particular emphasis on optimizing sample size, addressing class imbalance, and examining temporal effects within the framework of Geographically and Temporally Weighted Co-Location Quotient (GTWCLQ) analysis. We apply this approach to investigate the patterns of childhood respiratory diseases in Nanning City, using data at both monthly and daily scales from December 2016.

Results

By utilizing datasets spanning different time scales, we discern the spatio-temporal association patterns of childhood respiratory diseases and compare disparities across these temporal scales. Our findings reveal a higher aggregation of childhood respiratory diseases in Nanning City on a daily scale, particularly on days with poor air quality, compared to days with good air quality. Moreover, the experimental results show that temporal resolution can affect the intensity of the co-occurrence pattern, while duration influences its frequency, and starting time affects both intensity and frequency.

Conclusion

Our findings demonstrate the utility of this practical guide in managing sample size and class imbalance within GTWCLQ analysis, establishing it as a valuable tool for exploring multi-scale spatio-temporal co-location patterns. Furthermore, this study enhances our understanding of the spatio-temporal distribution of childhood respiratory diseases, providing insights that can aid in identifying and mitigating potential underlying causes, which is of considerable significance for GIS-based health analysis and decision-making.

1 Introduction

The increasing prevalence of respiratory diseases, driven by severe air pollution in numerous cities, has become a critical issue in achieving the objectives of the Healthy city [1, 2]. The effects of air pollution, including indoor contaminants, on children aged three and younger pose significant challenges for local environmental planning and management [3–5]. Typically, local hospitals record these diseases, noting the residence location and time of each incident. Investigating the spatial and temporal associations among various categories of childhood respiratory diseases is a crucial preliminary step before engaging in the modelling of demographic, environmental, and behavioral factors influencing these diseases. Such analysis provides valuable insights for targeted interventions and policy-making aimed at mitigating the impact on affected populations.

The Co-location Quotient (CLQ), originally formulated by Leslie et al. [6], is highly effective in identifying both symmetrical and asymmetrical spatial associations among categorical incidents. This approach has evolved into the geographically weighted CLQ (GWCLQ), introduced and developed by Cromley et al. [7] andWang et al. [8], and further refined into the geographically and temporally weighted CLQ (GTWCLQ), as presented by Li et al. [9]. These adaptations account for spatial and spatio-temporal heterogeneity, respectively. However, despite these advancements, CLQ and its derivatives—GWCLQ and GTWCLQ—continue to face challenges related to sample size, class imbalance, and temporal effects, particularly in spatio-temporal analyses. The proximity of incident data in both space and time often results in shared, unobserved effects, necessitating careful consideration of potential associations between incident categories to prevent biased statistical outcomes, as emphasised by Helai [10].

Previous research, including studies by Rahman et al. [11] and Johnson et al. [12], has demonstrated the significant influence of class imbalance and small sample sizes on statistical model parameter estimates. For instance, with small sample sizes, widely used metrics like Pearson's correlation coefficient, which is valuable for assessing geographical associations, can become unreliable due to interdependencies between observations, as noted by Choi et al. [13]. Similarly, logistic regression, a conventional method for analysing and modelling binary variables, is not immune to these issues. Cumming [14] showed how reducing sample sizes can adversely affect the accuracy of logistic regression models. Cheng et al. [15] introduced a method to achieve balanced sample sizes between binary variables for logistic regression modelling by employing systematic and random sampling. Additionally, Hu et al. [16] highlighted how class imbalances within logistic regression and other probabilistic models can bias results towards the more prevalent class, often neglecting the rarer one.

However, the impact of sample size and class imbalance has not been thoroughly examined in the context of co-location quotient analysis. Moreover, the selection of bandwidth has not been sufficiently justified. Bandwidth determines the influence range among adjacent observations, a key aspect in defining spatial relationships and scale [17, 18]. For accurate depiction of spatial patterns and relationships in spatial weight analysis, it is essential to choose a bandwidth that aligns with the data's spatial characteristics. Various methods exist for selecting a suitable bandwidth, ranging from visual analysis of spatial patterns to empirical approaches tailored to the data's unique properties, and statistical methods such as cross-validation or information criteria [19].

Notably, in specific spatial association pattern detection models, such as the cross-k-function [20] and co-location mining [21], the spatial relationship can vary with adjustments to the bandwidth. As a result, the outcomes of these analyses may differ when conducted at different spatial and temporal scales. The absence of a one-size-fits-all solution for determining the optimal bandwidth highlights the need for a thoughtful, data-specific approach to bandwidth selection in these analyses. It is imperative to consider the specific goals of the analysis and the unique characteristics of the spatial data being used. This strategic approach to bandwidth selection is vital to ensure the accuracy and relevance of results in spatial statistical analyses.

In this study, we present an innovative approach to examining incident data, aimed at addressing the research challenges outlined earlier. Our investigation is centred around three key questions:

1. How can we effectively tackle the issues of sample size and class imbalance in the analysis of categorical incident data?
2. What is the optimal approach for selecting an appropriate bandwidth for spatial and temporal scale analyses in categorical incident data?
3. How can we analyse association relationships across multiple temporal scales and distinguish variations among these different temporal dimensions?

To address these questions, the structure of the paper is as follows: The next section provides a detailed explanation of the Geographically and Temporally Weighted Co-Location Quotient (GTWCLQ) method, along with practical strategies for managing sample size and class imbalance. Section 3 focuses on applying this practical guide to explore the spatio-temporal patterns of childhood respiratory diseases in Nanning City. Finally, Section 4 summarises our findings and discusses potential avenues for future research.

2 Methodological Guide

2.1 Geographically and temporally weighted co-location quotient (GTWCLQ)

The GTWCLQ possesses the capability to analyse spatio-temporal associations, including co-location, asymmetric dependence, and independence among categorical variables across various scales. For a comprehensive exposition of the GTWCLQ's methodology and technical intricacies, we direct readers to our recently published work by Li et al. [9]. In this context, we specifically emphasise and elucidate its mathematical formulation, encompassing both global and local aspects. The global GTWCLQ analysis, designed to provide a thorough overview of spatio-temporal patterns, is expressed as follows (Eq. 1):

$$GTWCLQ_{A-B}^{\text{global}} = \frac{\sum\limits_{k\in E} N_k - 1}{\sum\limits_{k\in E} N_{Bk}} \times \frac{\sum\limits_{k\in E}\sum\limits_{i\in C}\sum\limits_{j\in D_k} W_{ijk}x_{kj}}{\sum\limits_{k\in E}\sum\limits_{i\in C}\sum\limits_{j\in D_k} W_{ijk}}$$

1

where $\mathrm{GTWCLQ}_{\mathrm{A-B}}^{\mathrm{global}}$ measures the global geographically and temporally weighted colocation quotient of type A relative to type B. $W_{ijk}$is a spatio-temporal weight matrix. $x_{kj}$is a binary value equal to 1 if the jth point at time k is a type B point, and is equal to 0 otherwise. $\mathrm{N_k}$ and $\mathrm{N_{Bk}}$ are the total number of points and type B points at time k, respectively. E is the set of time periods under study. C is the set of type A points at time k and D_k is the set of all points at time k.

The local GTWCLQ, which can examine the spatial heterogeneity of association patterns across the study area, is formulated as follows (Eq. 2):

$$GTWCLQ^{\text{local}}(i) = \frac{\sum\limits_{k\in E} N_k - 1}{\sum\limits_{k\in E} N_{Bk}} \times \frac{\sum\limits_{k\in E}\sum\limits_{j\in D_k} W_{ijk}x_{kj}}{\sum\limits_{k\in E}\sum\limits_{j\in D_k} W_{ijk}}$$

2

where $\mathrm{GTWCLQ}^{\mathrm{local}}(i)$ denotes the local value of ith type A relative to type B. The local GTWCLQ, which calculates a value at each point, rather than for the whole region, not only identifies whether a point pattern is more clustered or dispersed but also allows mapping and visualization to be

employed for interpretation such as LISA.

The GTWCLQ value describes the spatio-temporal colocation relationship between points of different categories. When the value is equal to 1, it indicates the random spatial association between categories, and a value greater than 1 indicates that type A tend to be spatially associated with type B points, while a value less than 1 indicates that type A points are likely to be far from type B. A higher value indicates a stronger dependence or attraction. Additionally, the definition of the GTWCLQ is unidirectional (i.e., $\mathrm{GTWCLQ}_{\mathrm{A-B}}^{\mathrm{global}}$ not necessarily equal to $\mathrm{GTWCLQ}_{\mathrm{B-A}}^{\mathrm{global}}$). If both the $\mathrm{GTWCLQ}_{\mathrm{A-B}}^{\mathrm{global}}$ and $\mathrm{GTWCLQ}_{\mathrm{B-A}}^{\mathrm{global}}$ are greater than 1 means symmetric association, while either of them less than 1 means asymmetric association.

A key feature of the GTWCLQ approach is the spatio-temporal weight matrix. The spatio-temporal weight matrix is constructed by multiplying the spatial weight matrix and the temporal weight matrix, term by term, as shown in Equations (3–5).

$$W_{ijk}=W_{ij}T_k=f(d_{ij}^s)\odot g(d_{ij}^t)$$

3

$$W_{ij}=f\left(d_{ij}^s\right)=\exp\left(-\frac{1}{2}\left(\frac{d_{ij}^s}{b_0}\right)^2\right)\quad if\quad d_{ij}^s\leq\ b_0$$

4

$$T_K=g\left(d_{ij}^t\right)={d_{ij}^t}^{-2}\ \ if\ \ d_{ij}^t\leq\ t_0$$

5

where d_{ij}^s and d_{ij}^t represent the spatial and temporal distance, respectively, f(.) and g(.) are the spatial and temporal weight kernel functions, respectively. t_0 and b_0 denote the temporal bandwidth and spatial bandwidth, respectively. The bandwidth is a crucial parameter in the calculation of spatio-temporal weight matrix, which corresponds to the spatial scale and temporal scale. Bandwidth selection can be either fixed or adaptive, with the following section providing a detailed discussion on this topic. A Monte Carlo simulation is employed to compare the observed GTWCLQ with the expected values under the null hypothesis, allowing for the determination of statistical significance (i.e. p-value) of the calculated values. All statistical tests are two-sided, with P < 0.05 considered statistically significant. The GTWCLQ software package developed by the team is available with the identifier at the following link https://doi.org/10.6084/m9.figshare.16641763. This study utilised Microsoft Office Excel for data statistics, ArcGIS Pro10.8 for data analysis and mapping.

2.2 The sample size of categorical incidents

The GTWCLQ was originally developed for analysing patterns in big data contexts. However, it is crucial to acknowledge that applying GTWCLQ to datasets with a small sample size can still yield significant results in co-location detection, regardless of the data's inherent characteristics. This situation may potentially lead to biased outcomes. Therefore, while GTWCLQ is a powerful tool for large datasets, caution must be exercised when dealing with smaller datasets to ensure the reliability and validity of the results.

As previously described, GTWCLQ relies on a hypothesis testing paradigm, specifically the significance test using the Monte Carlo method. The Monte Carlo method employs the z-test, which performs well with large sample sizes but is known to be less reliable with smaller sample sizes [22]. To ensure the detection of reliable spatial association patterns with GTWCLQ, it is necessary to impose a minimum sample size restriction.

The minimum sample size, taking into account various factors such as effect sizes, the number of observations, and standard deviation, has been computed using several methods [23].

Typically, for standard values of a two-sided alpha level of α = 5% (significance level) and a confidence interval width of E = 0.05, the minimal size n_{min} can be computed by Eq. (6) [24]:

$$\mathrm{n}_{min}=\left(\frac{Z_{\raisebox{1ex}{\$\alpha\ \$}/\raisebox{-1ex}{\$2\$}}*\sigma}{E}\right)^2$$

6

where σ is the standard deviation, Z is the significant level value related to the desired confidence interval E.

In addition to the complex formulae presented earlier (as in Eq. 6), some papers provide straightforward rules of thumb to assist researchers in determining the minimum sample size. For instance, VanVoorhis et al. [25]recommend that the minimum sample size for correlation or regression analyses should be no less than 50 + 8 in the social sciences. Tabachnick et al. [26] suggest a minimum of 300 cases or a more lenient guideline of

50 participants per factor for factor analysis. In practice, and as applied in our research, we have chosen a minimum sample size of 30, a threshold commonly used in various statistical methods, including multilevel Pearson's R [13], regression models [27], and Moran's I [28]. However, it is essential to exercise caution when dealing with category sample sizes, aside from the total sample size.

Leslie et al. [6] advises discontinuing the use of the CLQ model if the number of individuals in a category falls below 10. Sample sizes determined based on the precision of performance statistics, such as those in Eq. 7, typically result in larger sample size requirements compared to rules of thumb. If the sample size falls below this minimum threshold, the effectiveness of GTWCLQ may be compromised. In such cases, potential solutions include category aggregation or employing visualisation techniques within a GIS framework.

2.3 Class imbalance between categorical incidents

As demonstrated by Equations (1–2), the GTWCLQ value is notably sensitive to the proportion of category samples within the dataset. A significant proportion of category samples tends to result in a lower co-location value, as their neighbouring incidents occur less frequently than expected. Conversely, a smaller proportion of samples can yield a higher value, leading to an overestimation of spatial association. This challenge becomes particularly pronounced when dealing with same-category associations, especially in the presence of substantial differences in category sample sizes, a phenomenon known as class imbalance.

In practical applications, class imbalance is a prevalent issue across various domains and often leads to reduced accuracy for the minority class [29, 30]. Numerous methods exist to address the class imbalance problem, including under-sampling the majority class and oversampling the minority class during data preprocessing [31], as well as machine learning methods [32]. However, these techniques may not be suitable for GTWCLQ analysis, as resampling can disrupt the original distribution pattern of spatial data. To maintain the integrity of the categorical data distribution, we have chosen to address class imbalance in a statistical manner by merging or deleting small-sized categories. This approach allows us to balance the classification between categories while preserving the inherent distribution of the spatial data.

To assess the degree of class imbalance within the dataset, we employ a two-class imbalance ratio (IR) metric[33]. This metric is defined as the ratio of the number of instances in the majority class (referred to as the negative class) to the number of instances in the minority class (referred to as the positive class). In our multi-class analysis, we introduce a novel imbalance ratio (IR) index tailored to this context. The formula for this multi-class imbalance ratio (IR) is represented as follows in Eq. (7).

$$\text{IR}= \max_{A,B\in F}\left(\frac{N_A}{N_B}\right) \quad (7)$$

Where N_A and N_B represent the number of the type A and type B, respectively, and F is the set of categories. The IR value falls within the range of 1 to *N*-1, with a higher value indicating a more pronounced class imbalance. Generally, if the ratio exceeds 4, the dataset is considered as have class imbalance [34]. An IR value ranging from 4 to 10 indicates a moderate degree of class imbalance, whereas an imbalance ratio exceeding 100 indicates to a severe imbalance among the categories.

Furthermore, it is important to note that small sample sizes can exacerbate the issue of class imbalance[35, 36]. In large datasets (comprising 100,000 instances or more) [37], the class imbalance problem can be mitigated with low to moderate imbalance ratio values, but it remains highly sensitive to severe class imbalance[38, 39]. Consequently, we recommend considering the deletion of the minority class or merging it with an existing larger class that shares similar characteristics when the class ratio exceeds 100 in large datasets.

Additionally, for smaller datasets, which typically contain thousands to tens of thousands of instances, the performance of GTWCLQ tends to deteriorate with small sample sizes[40]. Depending on the imbalance ratio (IR), the following strategies are recommended for handling class imbalance within the GTWCLQ analysis:

1. IR < 4: In cases where the imbalance ratio is less than 4, the GTWCLQ analysis can be applied directly without the need for special adjustments, regardless of sample size considerations.
2. 4 ≤ IR < 10: When the dataset exhibits an imbalance ratio between 4 and 10, and includes one small-sized class alongside larger ones, the recommended approach is to merge the small-sized class into an existing larger class with similar characteristics. If there are multiple small-sized classes, it is advisable to merge these distinct categories into a super class. In situations where the dataset comprises one class with a large number of examples, while others contain only a few, distance-based under-sampling methods [41] or clustering-based under-sampling methods[42] can be employed to balance the class sizes.
3. IR ≥ 10: When the imbalance ratio exceeds 10, and the sample size is greater than 1,000, several options are available. These include deleting the minority class, merging the minority class into an existing larger class, or deciding not to employ the GTWCLQ method for the analysis.

These strategies are designed to address class imbalance across different scenarios, ensuring that GTWCLQ analysis yields reliable and meaningful results.

2.4 The selection of bandwidth in GTWCLQ

The selection of bandwidth is a critical aspect of GTWCLQ application, as it determines the spatial and temporal scale at which associations are assessed. Extensive literature has shown that co-location patterns can vary with different bandwidths, making the determination of the optimal bandwidth a challenging task [43, 44]. Consequently, the choice of bandwidth depends on prior domain knowledge, research objectives, and extensive experimentation.

In practical applications, the "Rule of Thumb" (ROT) is often used to estimate an initial bandwidth, and it has been shown to outperform first-generation methods for independent data[45]. The ROT serves as a valuable starting point for further analysis. In this study, we provide guidance on selecting an appropriate spatial and temporal bandwidth to assist researchers in making informed decisions regarding the scale of analysis.

(1) The spatial bandwidth selection

As is commonly understood, spatial bandwidth is typically constrained within a specific distance range and cannot be infinite in practical applications. To establish a range of values for a fixed bandwidth, we can define the neighbor distance range as follows:

Let the F is the category set $\mathrm{F} = [F_1, \mathrm{F}_2, \ldots, \mathrm{F}_n]$. Set G_i is the set of instances corresponding to category F_i, and G_k is the set of instances corresponding to category F_k, the g_i^j is the *jth* instance of G_i, and the nearest neighbor distance between e_i^j and G_k is define as

$$\mathrm{D}_k\left(g_i^j\right) = \min\left(\mathrm{dist}(\, g_i^j, \mathrm{G_k}\,)\right)$$

8

Where dist( ) commonly refers to the Euler distance. The neighbor distance range (NDR) can be defined as:

$$\mathrm{NDR} = (\min_{\mathrm{k}\in\ \mathrm{F},\mathrm{i}\in\ \mathrm{F},\mathrm{j}\in\ \mathrm{G_i}}\left(\mathrm{D}_k\left(g_i^j\right)\right), \max_{\mathrm{k}\in\ \mathrm{F},\mathrm{i}\in\ \mathrm{F},\mathrm{j}\in\ \mathrm{G_i}}\left(\mathrm{D}_k\left(g_i^j\right)\right))$$

9

The NDR represents the span of spatial bandwidth, providing an approximation of the spatial effect region for the points. Within this range, researchers can select the appropriate spatial bandwidth in line with their research objectives. However, it is important to note that determining the spatial bandwidth range can be computationally demanding, which presents a challenge for some researchers. Additionally, there are "Rule-of-Thumb" guidelines available for selecting the bandwidth in GTWCLQ. Specifically, if the kernel function f(·) is the Gaussian kernel (as shown in Eq. 4), then the bandwidth for GTWCLQ is often defined as shown in Eq. (10):

$$\mathrm{b} \approx\ 1.06\sigma\,\mathrm{N}^{-\frac{1}{5}}$$

10

where N is the number of data points and the parameter σ denotes the standard deviation of the distance between each pair of points [46]. For simplicity, the above equation can be condensed to b= $\mathrm{N}^{-\frac{1}{5}}$ when rough results are acceptable without accounting for the sample distribution. This formula is applied to random variables with a normal distribution. When dealing with data that does not conform to a normal distribution, the results may experience some degree of smoothing or deviation.

When deliberating on adaptive bandwidth and the optimal value of K-nearest neighbors, it's crucial to account for the range of bandwidth selection. Various studies have suggested setting the bandwidth to the square root of the number of data points, following the K-NN (K-nearest neighbors) rule [47].

$$\mathrm{K} = \sqrt{\mathrm{N}}$$

11

It is essential to emphasise that the maximum value for K-nearest neighbours cannot exceed the sample size of the smallest category. Surpassing this limit may prevent the method from finding a sufficient number of neighbours when computing values within the same category.

While the "Rule of Thumb" (ROT) may not always represent the optimal choice, it is used for its speed and reasonable estimation qualities. It often serves as an initial estimator in multi-stage bandwidth selection processes. Compared to fixed bandwidth, adaptive bandwidth offers the advantage of ensuring that the same number of points are involved in estimating the Co-Location Quotient (CLQ) at each marked point. This characteristic leads to more robust and reliable results. In practical applications, adaptive bandwidth is more commonly used, as it can effectively capture variations in spatial density across different local regions, making it a valuable tool for GTWCLQ analysis.

(2) The temporal bandwidth selection

The selection of temporal bandwidth can have a similar impact on statistical results as the choice of spatial bandwidth in analyses. However, while empirical suggestions exist for choosing spatial bandwidth in practical applications, there are no equivalent guidelines for selecting temporal bandwidth. The decision regarding temporal bandwidth typically depends on the specific research question, the characteristics of the data, and the available computational resources. It may require some trial and error or sensitivity analysis to determine the most suitable temporal bandwidth for a particular analysis [48, 49].

2.5 Temporal effects

The temporal effects encompass factors such as temporal resolution, duration (boundary), and starting time, which are measured by *t2-t1*, *tn-t1*, and *t1*, respectively [50]. In the realm of spatial analysis, researchers have extensively explored spatial scaling effects (e.g., Cheng and Fotheringham, 2013), while temporal effects have gained increasing attention within the domain of big data analytics [51, 52]. Temporal sensitivity can be refer to the intensity and frequency of individual events nested within environmental changes occurring at different timescales [53].

In the case of GTWCLQ analysis, accounting for temporal effects allows for a deeper understanding of the dynamic processes underlying co-location patterns. In this study, we have developed two indices to assess the frequency and intensity of significant colocation at various temporal and spatial scales, drawing inspiration from prior literature[54]. The colocation pattern frequency index D_f can be defined as follows:

$$V_{\text{frequency}} = mean\left(\sum_{\text{A}\in \text{F}} \sum_{B\in F} \frac{LN_{\text{A}-\text{B}}}{N_{\text{A}}}\right)$$

12

Where $V_{\text{frequency}}$ represents the average proportion of significant spatio-temporal colocation. The $LN_{\text{A}-\text{B}}$ denotes the number of $\text{GTWCLQ}^{\text{local}}$ values greater than 1 from type A to type B, and N_{A} is the number of type A. The index indicates the frequency of colocation pattern in the local GTWCLQ.

The colocation pattern intensity index $V_{intensity}$can be defined as follow:

$$V_{intensity} = mean\left(\sum_{A\in F} \sum_{B\in F} \frac{CLQ_{\text{A}-\text{B}}}{CLQ_{\text{A}}}\right)$$

13

Where $V_{intensity}$ denotes the average $\text{GTWCLQ}^{\text{local}}$ value with significant spatio-temporal colocation. The $CLQ_{\text{A}-\text{B}}$ is the value of $\text{GTWCLQ}^{\text{local}}$ greater than 1 from type A to type B, CLQ_{A} is the sum of $\text{GTWCLQ}^{\text{local}}$ value of type A. The index indicates the intensity of colocation pattern in the local GTWCLQ.

A larger value of $V_{\text{frequency}}$ indicates that a particular scale has a greater number of significant colocation points, making it more frequent than other scales. Conversely, a larger value of $V_{intensity}$ suggests that a scale exhibits more intense co-location patterns in terms of their spatial relationships compared to other scales.

3 Empirical case study

3.1 Study area and data sets

Acute respiratory diseases (ARD) are prevalent communicable illnesses, primarily attributed to widespread exposure to harmful inhalational factors in the environment, workplace, and personal behaviours [55, 56]. In 2019, China reported that 1.1 million children under the age of 5, constituting 10% of this age group, were diagnosed with ARD (GBD, 2019). Notably, children are more vulnerable to the effects of air pollution due to their higher breathing rates, narrower airways, developing lungs and immune systems, and increased outdoor exposure[3, 57]. Earlier studies have shown that the hospitalisation rate for acute respiratory illnesses in two-year-olds is approximately 12 times greater than in children aged 5–17 years without underlying high-risk conditions[58]. The primary focus of this research is on children aged 3 and under, a specific age group characterised by limited mobility confined to their residential neighbourhoods, in contrast to school students and adults.

This study was conducted in Nanning City [59], which serves as the capital of the less-developed Guangxi Zhuang Autonomous Region (GZAR). In recent years, the air quality in Nanning has declined due to urban development and increasing emissions from motor vehicles. Considering the distribution of pollution sources, the study area is confined to Nanning's five traditional urban districts, excluding the five counties that fall under its jurisdiction. The location of the study area is illustrated in Fig. 1.

The daily data concerning childhood respiratory diseases were sourced from the Maternal and Children's Health Hospital of Guangxi Province, covering the period from 1st January to 31st December 2016. The dataset includes information on gender, age, residential address, admission date, and a brief description of disease symptoms. It is important to note that all epidemiological data used in this study were ethically approved solely

for academic research purposes, with strict measures in place to safeguard patient privacy by excluding any personal identifiers, such as names and ID card numbers, before releasing the data. Based on diagnoses provided by medical professionals, the childhood respiratory diseases were categorised as follows: Acute nasopharyngitis (C1), Breathing pneumonia (C2), Acute pharyngitis (C3), Bronchitis (C4), and Polyangiitis-bronchitis (C5).

To ensure the accuracy of the information, this study underwent several data preprocessing steps. Initially, records lacking addresses or originating from areas outside the urban region, as well as those associated with individuals older than 3 years, were systematically removed. Secondly, a geo-coding technique [60] was employed to convert the residential addresses of patients into locational coordinates, represented as latitude and longitude. The geo-coding process applied to the current dataset resulted in a matching rate of 80%. After these meticulous data processing and filtering steps, the study included a total of 131,619 incidents involving children aged 3 and below residing in the five districts of Nanning in 2016. Each incident was represented as a point vector, denoted as Pi = (Location, Time, Disease category). The spatial distribution of these categorical incidents throughout the urban area of Nanning is illustrated in Fig. 2.

To facilitate the analysis of monthly patterns, the daily data were temporally aggregated on a monthly basis. The statistical characteristics of the disease dataset are illustrated in Fig. 3. Firstly, it is evident that the dataset suffers from a class imbalance issue among the five disease categories. As shown in Fig. 3 (left), there is a significant disparity in the total number of incidents across the five categories. Acute nasopharyngitis (C1) accounts for more than 50% of cases, while Polyangiitis-bronchitis (C5) represents only 1.28% of the cases. Secondly, the data exhibit pronounced temporal variations throughout the year 2016. As indicated in Fig. 3 (right), the highest incidence of respiratory diseases occurred during the spring months (March, April, May) and winter months (November, December, January), coinciding with periods of rapid temperature changes.

The air pollutant data were gathered from the China National Environmental Monitoring Centre (CNEMC)'s national real-time municipal air quality platform. PM 2.5 (Particulate Matter with a diameter of 2.5 micrometres) has been shown to be closely associated with childhood respiratory diseases[61, 62]. In our study, we selected December as the focus for monthly pattern analysis due to the recorded poorest air quality in 2016, as depicted in the left graph of Fig. 4. Additionally, we chose two specific days, the 19th and the 27th, for analysing daily patterns under varying air conditions. Figure 4 (right) illustrates that the 19th represents the single day with the highest PM 2.5 levels, while the 27th exhibits the lowest PM 2.5 levels.

3.2 GTWCLQ analysis

To validate the proposed approach, which addresses issues related to sample size, class imbalance, and temporal effects, we conducted an experiment to examine the spatio-temporal patterns of disease incidents and identify high-risk co-locations. This is crucial for effective disease management, and we utilised the GTWCLQ tool for this purpose.

Initially, we employed the GTWCLQ approach to assess both global and local spatial associations among the five types of diseases on a monthly scale. Next, we conducted a more detailed analysis to investigate spatial association patterns on a daily scale, taking into account different air conditions. Finally, we analysed the temporal effects across multiple temporal scales.

To comprehensively understand the spatial patterns at different temporal scales, we utilised various datasets to calculate the GTWCLQ, as outlined in Table 1. Using Equations (6) and (7), we determined the minimum sample size and imbalance ratios for each dataset. Specifically, the "GTWCLQ12" dataset was used for monthly pattern analysis, while "GTWCLQ31" was employed for daily pattern analysis in December. Additionally, the "GTWCLQ7" dataset was utilised for daily pattern analysis under different air conditions.

As shown in Table 1, all datasets met the minimum sample size requirements, with the first dataset considered a large dataset (e.g., containing more than 100,000 instances), while the remaining datasets were categorised as small datasets, containing thousands to tens of thousands of instances. Moreover, all datasets exhibited varying degrees of class imbalance issues. In the following section, we will demonstrate the application of the GTWCLQ approach on these different datasets with varying imbalance ratios.

Table 1
Datasets processed for GWTCLQ analyses

Dataset	Temporal scale	Sample size	Scale (resolution and duration)	n_{min}	IR
GTWCLQ12	12 months	131,619	Monthly, Jan-Dec	383	34
GTWCLQ31	31 days in Dec	10,674	Daily, 1-31st	370	29
GTWCLQ7	Seven days in Dec	1,064	Daily, 13-19th, 20-27th	282	20

All experiments were conducted on a computer equipped with 32GB of RAM, an Intel Core i7 CPU, and running the Windows 10 operating system.

3.3 GWTCLQ12: Monthly spatial-temporal pattern in Nanning

The dataset "GTWCLQ12", comprising 12 months and a total of 131,619 records was utilised for monthly pattern analysis. The imbalance ratio for "GTWCLQ12" was calculated as 34, determined by comparing the majority class C1 with the minority class C5 (IR = $^{57314}/_{1688} \approx 34$) using formula (8). It's important to note that "GTWCLQ12" is considered a large dataset, exceeding 100,000 instances.

As detailed in section 2.2.3, the GTWCLQ method can be applied directly to this dataset without the need to address the class imbalance problem. Initially, December was selected as the current time frame for studying the monthly spatio-temporal pattern in 2016, resulting in a temporal bandwidth set to 12 (months), and the temporal period designated as "month." Furthermore, considering the uneven distribution of data across different areas, an adaptive spatial bandwidth with a K-nearest value of 362 ($\mathrm{K} = \sqrt{131619} \approx 362$) was chosen, following formula (11).

Additionally, the sampling size for Monte Carlo significance test was set at 1000, with the number of simulations typically established at 1,000, at a significance level of 0.05, as per the methodology outlined in previous literature [63, 64]. With these parameter settings in place, the global GWTCLQ results for all five categories are presented in Table 2 and Fig. 5.

Table 2
The global values of GTWCLQ12 in monthly pattern

Category	C1	C2	C3	C4	C5	Sample size
C1	0.957	***1.079***	0.995	0.908	***1.076***	57314
C2	0.856	***1.189***	0.943	0.950	***1.199***	48569
C3	0.880	***1.049***	***1.659***	0.914	***1.147***	8544
C4	0.811	***1.090***	0.929	***1.441***	***1.140***	15504
C5	0.804	***1.123***	0.997	***1.036***	***3.791***	1688
Note: all significant at 5% level						

Table 2 provides insights into the spatial patterns of disease categories. Specifically, four disease categories (C2–C5) exhibit same-category CLQ values greater than 1, which are highlighted in bold and italic in the table. These values along the diagonal indicate that these disease categories have a strong spatial autocorrelation, taking into account both spatial and temporal distance decay effects. However, Acute nasopharyngitis (C1) shows a random distribution, as its CLQ value is less than 1. Among all the diseases, Polyangiitis-bronchitis (C5) exhibits the most significant clustering pattern with itself, as evidenced by the highest CLQ value of 3.791.

Furthermore, the global GWTCLQ analysis detects symmetry and asymmetry in dependence between the five categories, as depicted in Fig. 5. Two pairs of categories exhibit symmetrical association or co-location: Polyangiitis-bronchitis (C5) and Breathing pneumonia (C2), as well as Polyangiitis-bronchitis (C5) and Bronchitis (C4). This indicates that these two diseases tend to co-occur in the same locations. Conversely, there are asymmetric association between C1 and C2, C3 and C2, and C4 and C2. Specifically, Breathing pneumonia (C2) and Polyangiitis-bronchitis (C5) are the most dependent categories[59], as all other categories have GTWCLQ values greater than 1 in relation to them, but not vice versa. This suggests that Breathing pneumonia (C2) and Polyangiitis-bronchitis (C5) tend to attract many other diseases to occur in their vicinity.

The global GTWCLQ provides an overall assessment of spatial associations between different disease types, while the local GTWCLQ enables a more granular analysis of spatio-temporal associations from one location to another. Figure 6 illustrates the local GTWCLQ values, highlighting associations between each disease and other disease types. Only local GTWCLQ values greater than one and statistically significant ($p < 0.05$) are considered, as these provide valuable insights for understanding the distribution patterns of diseases.

Firstly, from a spatial autocorrelation perspective, C5 and C3 exhibit different distribution patterns. As depicted in Fig. 6a, Polyangiitis-bronchitis (C5) is primarily concentrated in the city centre, where population and traffic density are higher compared to other areas. In contrast, Acute pharyngitis (C3) is prevalent on both sides of the river, particularly in the southern region (Fig. 6b).

Secondly, the spatio-temporal association with other diseases varies for each type. As mentioned earlier, all other disease types display asymmetric association with Polyangiitis-bronchitis (C5) in the city centre and residential areas, indicating that these diseases tend to cluster with the flow of people (Fig. 6c). Additionally, there is a symmetrical association between C2 and C5, as shown in Fig. 6d, with different patterns of high concentration. Notably, the inter-attraction association mainly occurs in densely populated areas, such as around the railway station (Fig. 6c). This could be attributed to higher levels of air pollution in these areas, which are more likely to trigger the occurrence of these two diseases.

3.4 GTWCLQ31: Daily spatial-temporal pattern in Nanning

The study also included the "GWTCLQ31" dataset, which covered 31 days in December 2016, to examine the daily spatio-temporal pattern in Nanning. Given this specific time frame, a temporal bandwidth of 31 days was chosen, with the temporal periods set as "day". However, due to the traditional nature of the "GWTCLQ31" dataset and an imbalance ratio exceeding 10, specifically 29, it was deemed inappropriate to directly apply the GTWCLQ model, as outlined in the guidelines in section 2.2.

To address this issue, given the limited sample size of class C5 in December, totalling 151 cases with only one case on the 17th, and its small proportion (only 1.4%), class C5 was merged into C3, as both exhibit similar symptoms. By combining C3 and C5, the imbalance ratio (IR) of "GWTCLQ31" was reduced to 5, meeting the conditions for statistical analysis. Additionally, an adaptive spatial bandwidth was applied to account for the uneven distribution of diseases. Conventionally, the K value could be set to 100 based on formula (12). However, considering the limited number of diseases on the 11th (only 54 cases) and taking into account the randomness of the Monte Carlo sampling, this study opted for a K value of 25, which is half the size of the smallest sample observed on the 11th. The simulation sample size was set to 50, and the simulation iterations were set to 1000 for Monte Carlo analysis.

Following the merging of classes and the specified parameter settings, the global GTWCLQ values at a significance level of 0.05, across all four categories, are presented in Table 3.

Table 3
Global values of GWTCLQ31 at daily level(time scale is 31 days)

Category	C1	C2	C3	C4	Sample size
C1	1.004	1.032	0.690	1.101	4200
C2	0.723	1.305	0.420	1.286	4396
C3	0.488	1.030	2.670	1.453	889
C4	0.664	0.773	1.127	2.938	1187
Note: all significant at 5% level					

The local co-location values from GWTCLQ31 at daily level are depicted in Fig. 7.Compared to the monthly pattern, the daily pattern observed in December exhibits stronger clustering within the same category. Among the disease categories, Bronchitis (C4) shows the most pronounced clustering within its category in the daily pattern (Fig. 7a-b), indicating a higher tendency for cases of Bronchitis to cluster together. In contrast, Acute Nasopharyngitis (C1) displays relatively weaker clustering within its category. Additionally, symmetrical associations exist between certain categories. Specifically, Acute Nasopharyngitis (C1) is co-located with C2 and C4, but the inverse is not true. This suggests that most cases of Acute Nasopharyngitis are likely caused by localised disease outbreaks rather than transmission from other diseases. Furthermore, all diseases exhibit co-location with C4 (Fig. 7c), indicating that Bronchitis is consistently associated with all childhood respiratory diseases.

Moreover, a symmetrical co-location relationship was identified between Acute Pharyngitis (C3) and Bronchitis (C4), suggesting that these two diseases frequently cluster at the same locations and times (Fig. 7d). These findings support the relevance of classifying the pathogenesis of these diseases and confirm their spatio-temporal associations.

3.5 GWTCLQ7: Daily spatial-temporal pattern in different air condition

The association between respiratory diseases and air quality in residential neighbourhoods is well-established. Using the GWTCLQ method, various spatio-temporal patterns can be explored under different air quality conditions. For this analysis, two specific days in December were chosen to represent contrasting air quality: the 19th, characterised by severe pollution, and the 27th, characterised by good air quality.

The "GWTCLQ7" dataset used in this experiment is small and suffers from a severe class imbalance (IR of 20). To address this, category C5 was merged into C3 based on disease characteristics. Previous studies have shown that respiratory diseases typically persist for 7 to 14 days[65]. Therefore, a temporal bandwidth of 7 days was selected, reflecting the course of the disease. Considering the impact area of air pollution, a fixed bandwidth was chosen for this experiment. Some studies have suggested that the influential buffer radius of air pollution ranges from 100 to 3,000 metres[66]. To assess the extent of pollution in Nanning, the neighbouring distance range in the "GWTCLQ7" dataset was calculated using formulas (8–9), and found to range from 1,000 to 3,000 metres, with a mean distance of 2,000 metres. Consequently, 2,000 metres was selected as the spatial bandwidth for this study. Additionally, the box kernel function was chosen as the spatial kernel function due to the nature of air pollution. Monte Carlo simulations were performed 1,000 times, with the simulation sample size set to 25. The results of the global GWTCLQ values and the local distribution are presented in Table 4 and Fig. 8.

The analysis of spatial association on a 7-day temporal scale revealed weak spatial associations between air conditions, with notable differences observed between the two. Among the same disease categories, three types of diseases exhibited clustering on polluted days, except for C1. Conversely, on clean days, only C3 displayed a clustering pattern within the same category. This suggests that most childhood respiratory diseases tended to cluster under poorer air conditions. Regarding associations between different disease categories, C1 and C2 showed symmetric co-location under both air conditions. However, C1 and C4, as well as C1 and C3, exhibited inter-attraction on polluted and clean days, respectively. This indicates that these diseases often cluster at the same place and time. Additionally, all diseases were attracted to Acute Nasop haryngitis (C2) on polluted days, indicating that Breathing Pneumonia (C2) tends to attract other diseases in its vicinity.

Table 4
The global values of GWTCLQ7 in different air conditions (7-day time scale)

Day	Category	C1	C2	C3	C4
19th (max PM2.5)	C1	0.9248	**1.0670**	0.8692	**1.0094**
	C2	**1.0245**	**1.0173**	0.7990	0.9950
	C3	**1.0502**	**1.0289**	**1.0264**	0.7915
	C4	**1.0172**	**1.0317**	0.7064	**1.0111**
27th (min PM2.5)	C1	0.9592	**1.0311**	**1.1054**	0.9460
	C2	**1.0180**	**1.0033**	**1.0026**	0.9452
	C3	**1.0669**	0.9460	0.9156	**1.0589**
	C4	**1.0402**	**1.0382**	**1.0483**	0.7538

To illustrate the disparity between the two air quality conditions, the local co-location values of C1 and C2 under different air conditions are depicted in Fig. 8. It is clear that the symmetric co-location between Acute Nasopharyngitis (C1) and Breathing Pneumonia (C2) clusters more prominently on polluted days compared to clean days.

On the 19th, the clusters are primarily concentrated in the centre of Qinxiu and in rural areas of the Xingning district. These areas have a high density of the catering industry or the burning of straw, both contributing to higher pollution levels. In contrast, the cluster points on the 27th are less concentrated and more scattered, indicating that Acute Nasopharyngitis (C1) and Breathing Pneumonia (C2) in a clean environment exhibit a more random distribution pattern.

3.6 Temporal effects

The scaling effect has long been a central topic in Geographic Information Systems (GIS), with most studies primarily focused on spatial scales. However, the availability of big data, particularly high-velocity temporal data, has increasingly addressed concerns related to temporal effects. This is particularly important for GWTCLQ analysis. In this case study, the temporal scales are determined by both resolution (daily or monthly) and duration (one year or one month). The starting time effect refers to the sensitivity of the initial time to the spatio-temporal patterns.

To capture these temporal effects, including scale, duration, and starting time, two indicators—the frequency and intensity of colocation patterns (as defined in Equations 12–13)—were calculated and are presented in Table 5. These indicators help analyse the impact of various temporal factors on the observed spatio-temporal patterns.

Table 5
The frequency and intensity values between four scenarios

Index	GWTCLQ12	GWTCLQ31	GWTCLQ7(19th)	GWTCLQ7(27th)
$\varvec{V}_{\mathbf{frequency}}$	0.456	0.364	0.462	0.495
$\varvec{V}_{\varvec{intensity}}$	0.661	0.727	0.703	0.667
Resolution	Month	Day	Day	Day
Duration	1 year	1 month	1 week	1 week
Starting	January	1st Dec	13th Dec	21st Dec

Table 5 provides a comprehensive overview of the varying intensity and frequency of spatio-temporal associations across four scenarios. The key observations are as follows:

Frequency of Aggregation

The highest aggregation frequency is observed at the 7-day scale (GWTCLQ7s), followed by the monthly scale (GWTCLQ12), with the lowest frequency found at the daily scale (GWTCLQ31). This indicates that different types of respiratory diseases exhibit more pronounced spatial and temporal associations at the 7-day (one-week) scale (GWTCLQ7s).

Intensity Index

The daily scale generally demonstrates stronger intensity than the monthly scale (GWTCLQ12), and a longer duration (GWTCLQ31) results in higher intensity in the daily pattern.

Starting Day Effects

At the same temporal scales (resolution and duration), both frequency and intensity values vary with the starting day. The polluted day GWTCLQ7 (19th) shows intensive co-location but lower frequency, while the clean day GWTCLQ7 (27th) demonstrates less intensity but higher frequency.

Overall, this analysis reveals that temporal effects significantly influence the frequency of co-location, with the monthly pattern frequency being greater than the daily pattern. Additionally, temporal duration has a substantial impact on the intensity of co-location relationships, as the intensity over a year is less than that over a month. Furthermore, the starting time also affects both the frequency and intensity index, highlighting the importance of considering starting time effects in spatio-temporal analyses.

4 Conclusion

Categorical incident data is a common form of spatio-temporal data in urban GIS. In this paper, we developed a practical approach for exploring spatio-temporal associations between categories within incident data using Geographically and Temporally Weighted Co-Location Quotient (GTWCLQ) analysis. This study primarily addresses the challenges of minimum sample size and class imbalance when applying the GTWCLQ model. Key points include:

Minimum Sample Size: Determining the minimum sample size requires consideration of research objectives, background knowledge, and data characteristics such as variance. The study recommends an empirical minimum sample size of no fewer than 30 to ensure reliable results.

Class Imbalance Problem: Class imbalance is a common issue in categorical incident data analysis. The study proposes a class imbalance index, integrating it with sample size to guide the application of the GWCLQ method. Mitigating class imbalance is achieved through processes such as merging, removing, or resampling. Experiments with datasets exhibiting varying class imbalance indices demonstrate the effectiveness of this approach in addressing class imbalance.

Spatial Bandwidth Selection: The choice of spatial bandwidth significantly influences spatial effects. While previous studies have focused on identifying the optimal bandwidth, this study emphasises that there is no single optimal bandwidth for exploratory studies. Instead, researchers should select a bandwidth within a reasonable range. The study introduces a method for calculating the bandwidth range and provides various approaches for establishing fixed and adaptive bandwidths. These methods can be based on mathematical formulas (e.g., Formulas 8–9) or rule-of-thumb guidelines, offering a reference for researchers in determining suitable bandwidths. Experiments demonstrate how to set appropriate bandwidths in different scenarios, confirming the suitability of the proposed method.

Theoretical Foundation: The proposed bandwidth range method in this study can serve as a theoretical foundation not only for GTWCLQ but also for establishing the bandwidth for other spatial models, such as Moran's I. It provides valuable guidance for researchers in selecting spatial bandwidths effectively.

Temporal effects have long been a central focus and challenge in geographical research. Previous studies have predominantly focused on investigating temporal scales, often neglecting the duration and starting time of events, which can lead to inaccuracies. In this study, we propose a comprehensive methodological framework that accounts for temporal effects when analysing the distribution of event data under varying time conditions. We illustrate the analysis of spatio-temporal association patterns across multiple temporal scales using GTWCLQ and compare variations in these patterns at different scales. By selecting data at various temporal scales and controlling for temporal bandwidth, we examine the spatio-temporal association patterns of paediatric respiratory disease under different time effects. Additionally, we introduce two statistical indices to quantify the significant co-location frequency and strength within the local GTWCLQ. Our findings suggest that the multi-scale temporal effect framework based on GTWCLQ is valuable for exploring association patterns of event data at different temporal scales. Our framework will continue to evolve to address temporal effect issues across various fields and guide data selection for spatio-temporal data analysis.

In our research on respiratory diseases, we explore the patterns of childhood respiratory diseases at varying temporal scales in Nanning City. By utilising datasets spanning different time scales, we discern the spatio-temporal association patterns of childhood respiratory diseases and compare disparities across these temporal scales. Our findings reveal a higher aggregation of childhood respiratory diseases in Nanning City on a daily scale, particularly on days with poor air quality, compared to days with good air quality. Moreover, the experimental results show that temporal resolution can affect the intensity of the co-occurrence pattern, while duration influences its frequency, and starting time affects both intensity and frequency. These findings underscore the significance of temporal effects—namely temporal resolution, duration, and starting time—in influencing study outcomes. Therefore, in disease research, it is crucial to conduct comprehensive analyses and comparisons when selecting study data, ensuring the appropriate resolution, duration, and onset time. Failure to do so may result in inaccurate or misleading conclusions.

This study offers a fresh perspective by examining the spatio-temporal association patterns of incident data at different time scales. However, it also faces some limitations. The quantification of spatial association differences at different temporal scales remains unresolved. This study

provides some insights, primarily inferring the differences statistically, but the variations in spatial association remain an area for further research.

Abbreviations

GTWCLQ Geographically and Temporally Weighted Co-Location Quotient

GWCLQ Geographically Weighted Co-Location Quotient

CLQ Co-Location Quotient

IR Multi-class imbalance ratio

Declarations

Supplementary Information

The dataset and code supporting the conclusions of this article are available in the following link: https://doi.org/10.6084/m9.figshare.25037399.

Author contributions

LL: conceptual and methodological design, technical implementation, validation of the approach, interpretation of the results, writing, and revision. JC: conceptual and methodological design, writing and revision, and supervision.MX: technical implementation, software development, and revision. LD: data processing, writing, and revision. LM: validation of the approach, interpretation of the results, writing, and revision. JT: data collection, ethical approval, validation of the approach, interpretation of the results, writing, and revision.

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Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

References

1. Liu Y, Zhou Y, Lu J: **Exploring the relationship between air pollution and meteorological conditions in China under environmental governance**. *Sci Rep* 2020, **10**(1):14518.
2. Jadsri S, Singhasivanon P, Kaewkungwal J, Sithiprasasna R, Siriruttanapruk S, Konchom S: **Spatio-temporal effects of estimated pollutants released from an industrial estate on the occurrence of respiratory disease in Maptaphut Municipality, Thailand**. *International journal of health geographics* 2006, **5**(1):48.
3. Ibrahim MF, Hod R, Nawi AM, Sahani M: **Association between ambient air pollution and childhood respiratory diseases in low- and middle-income Asian countries: A systematic review**. *Atmos Environ* 2021, **256**:118422.
4. Jochem WC, Razzaque A, Root ED: **Effects of health intervention programs and arsenic exposure on child mortality from acute lower respiratory infections in rural Bangladesh**. *International journal of health geographics* 2016, **15**(1):32.
5. Hoy A, Mohan G, Nolan A: **An investigation of inequalities in exposure to PM2.5 air pollution across small areas in Ireland**. *International journal of health geographics* 2024, **23**(1):17.
6. Leslie TF, Kronenfeld BJ: **The Colocation Quotient: A New Measure of Spatial Association Between Categorical Subsets of Points.** . *Geogr Anal* 2011, **43**(3):306-326.
7. Cromley RG, Hanink DM, Bentley GC: **Geographically Weighted Colocation Quotients: Specification and Application**. *Professional Geographer* 2014, **66**(1):138-148.

8. Wang F, Hu Y, Wang S, Li X: **Local Indicator of Colocation Quotient with a Statistical Significance Test: Examining Spatial Association of Crime and Facilities**. *Prof Geogr* 2016, **69**(1):22-31.
9. Li L, Cheng J, Bannister J, Mai X: **Geographically and temporally weighted co-location quotient: an analysis of spatiotemporal crime patterns in greater Manchester**. *Int J Geogr Inf Sci* 2022(5/6):36.
10. Helai H: **Bayesian hierarchical analysis on crash prediction models**. 2008.
11. Rahman HAA, Wah YB, Huat OS: **Predictive Performance of Logistic Regression for Imbalanced Data with Categorical Covariate**. *Pertanika Journal of Science & Technology* 2021, **29**(1):181-197.
12. Johnson JM, Khoshgoftaar TM: **Survey on deep learning with class imbalance**. *J Big Data-Ger* 2019, **6**:1-54.
13. Choi J, Peters M, Mueller RO: **Correlational analysis of ordinal data: from Pearson's r to Bayesian polychoric correlation**. *Asia Pac Educ Rev* 2010, **11**(4):459-466.
14. Cumming GS: **Using between-model comparisons to fine-tune linear models of species ranges**. *J Biogeogr* 2000, **27**(2):441-455.
15. Cheng J, Masser I: **Urban growth pattern modeling A case study of Wuhan City, PR China**. In., vol. 62; 2013.
16. Hu JL, Tang XW, Qiu JN: **Analysis of the Influences of Sampling Bias and Class Imbalance on Performances of Probabilistic Liquefaction Models**. *Int J Geomech* 2017, **17**(6):1-13.
17. Fotheringham AS, Yu H, Wolf LJ, Oshan TM, Li Z: **On the notion of 'bandwidth' in geographically weighted regression models of spatially varying processes**. *Int J Geogr Inf Sci* 2022, **36**(8):1485-1502.
18. Lemke D, Mattauch V, Heidinger O, Pebesma E, Hense H-W: **Comparing adaptive and fixed bandwidth-based kernel density estimates in spatial cancer epidemiology**. *International journal of health geographics* 2015, **14**(1):15.
19. Li Z, Fotheringham AS, Oshan TM, Wolf LJ: **Measuring Bandwidth Uncertainty in Multiscale Geographically Weighted Regression Using Akaike Weights**. *Ann Am Assoc Geogr* 2020, **110**(5):1500-1520.
20. Noel, Cressie: **STATISTICS FOR SPATIAL DATA**. *Terra Nova* 1992, **4**(5):613-617.
21. Yao X, Chen L, Peng L, Chi T: **A co-location pattern-mining algorithm with a density-weighted distance thresholding consideration**. *Inform Sciences* 2017, **396**:144-161.
22. Muthén LK, Muthén BO: **How to use a Monte Carlo study to decide on sample size and determine power**. *Struct Equ Modeling* 2002, **9**(4):599-620.
23. Lakens D: **Sample Size Justification**. *Collabra-Psychol* 2022, **8**(1).
24. Kelley K: **Confidence Intervals for Standardized Effect Sizes: Theory, Application, and Implementation**. *J Stat Softw* 2007, **20**(8):1 - 24.
25. VanVoorhis CW, Morgan BL: **Understanding power and rules of thumb for determining sample sizes**. *Tutorials in quantitative methods for psychology* 2007, **3**(2):43-50.
26. Tabachnick BG, Fidell LS, Ullman JB: **Using multivariate statistics**, vol. 6: pearson Boston, MA; 2013.
27. Stegmueller D: **How many countries for multilevel modeling? A comparison of frequentist and Bayesian approaches**. *Am J Polit Sci* 2013, **57**(3):748-761.
28. Carrijo TB, da Silva AR: **Modified Moran's I for Small Samples**. *Geogr Anal* 2017, **49**(4):451-467.
29. Santos MS, Abreu PH, Japkowicz N, Fernández A, Soares C, Wilk S, Santos J: **On the joint-effect of class imbalance and overlap: a critical review**. *Artificial Intelligence Review* 2022, **55**(8):6207-6275.
30. van den Goorbergh R, van Smeden M, Timmerman D, Van Calster B: **The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression**. *J Am Med Inform Assn* 2022, **29**(9):1525-1534.
31. Huang C, Li Y, Loy CC, Tang X: **Deep Imbalanced Learning for Face Recognition and Attribute Prediction**. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 2020, **42**(11):2781-2794.
32. Huang C, Li Y, Chen CL, Tang X: **Deep Imbalanced Learning for Face Recognition and Attribute Prediction**. 2018(11).
33. Orriols-Puig A, Bernadó-Mansilla E: **Evolutionary rule-based systems for imbalanced data sets**. *Soft Computing* 2008, **13**(3):213-225.
34. Song Q, Guo Y, Shepperd M: **A Comprehensive Investigation of the Role of Imbalanced Learning for Software Defect Prediction**. *Ieee T Software Eng* 2019, **45**(12):1253-1269.
35. Branco P, Torgo L, Ribeiro RP: **A survey of predictive modeling on imbalanced domains**. *Acm Comput Surv* 2016, **49**(2):1-50.
36. Wasikowski M, Chen X-w: **Combating the small sample class imbalance problem using feature selection**. *Ieee T Knowl Data En* 2009, **22**(10):1388-1400.
37. Leevy JL, Khoshgoftaar TM, Bauder RA, Seliya N: **A survey on addressing high-class imbalance in big data**. *J Big Data-Ger* 2018, **5**(1).
38. Barua S, Islam MM, Yao X, Murase K: **MWMOTE–Majority Weighted Minority Oversampling Technique for Imbalanced Data Set Learning**. *Ieee T Knowl Data En* 2013, **26**(2):405-425.
39. Ghorbani M, Kazi A, Baghshah MS, Rabiee HR, Navab N: **Ra-gcn: Graph convolutional network for disease prediction problems with imbalanced data**. *Med Image Anal* 2022, **75**:102272.

40. Guan H, Zhang Y, Xian M, Cheng H-D, Tang X: **SMOTE-WENN: Solving class imbalance and small sample problems by oversampling and distance scaling**. *Appl Intell* 2021, **51**(3):1394-1409.
41. Kang Q, Lei S, Zhou M, Wang X, Wei Z: **A Distance-Based Weighted Undersampling Scheme for Support Vector Machines and its Application to Imbalanced Classification**. *Ieee T Neur Net Lear* 2018, **29**(9):4152-4165.
42. Lin W-C, Tsai C-F, Hu Y-H, Jhang J-S: **Clustering-based undersampling in class-imbalanced data**. *Inform Sciences* 2017, **409-410**:17-26.
43. Wang Z, Ye W, Chen X, Li Y, Zhang L, Li F, Yao N, Gao C, Wang P, Yi D *et al*: **Spatio-temporal pattern, matching level and prediction of ageing and medical resources in China**. *BMC Public Health* 2023, **23**(1).
44. Wang J, Cao W: **A Novel Approach for Mining Spatiotemporal Explicit and Implicit Information in Multiscale Spatiotemporal Data**. *Isprs Int J Geo-Inf* 2023, **12**(7):261.
45. Jones MC, Marron JS, Sheather SJ: **A brief survey of bandwidth selection for density estimation**. *J Am Stat Assoc* 1996, **91**(433):401-407.
46. Dharmani B: **Gram-Charlier A Series Based Extended Rule-of-Thumb for Bandwidth Selection in Univariate Kernel Density Estimation**. *Austrian Journal of Statistics* 2022, **51**:141-163.
47. Yuan K, Cheng X, Gui Z, Li F, Wu H: **A quad-tree-based fast and adaptive Kernel Density Estimation algorithm for heat-map generation**. *Int J Geogr Inf Sci* 2019, **33**(12):2455-2476.
48. Chang H, Praskievicz S, Parandvash H: **Sensitivity of Urban Water Consumption to Weather and Climate Variability at Multiple Temporal Scales: The Case of Portland, Oregon**. *International Journal of Geospatial and Environmental Research* 2014, **1**(1).
49. Wambura FJ: **Sensitivity of the Evapotranspiration Deficit Index to Its Parameters and Different Temporal Scales**. *Hydrology-Basel* 2021, **8**(1):26.
50. Cheng T, Adepeju M: **Modifiable Temporal Unit Problem (MTUP) and Its Effect on Space-Time Cluster Detection**. *PloS one* 2014, **9**(6):e100465.
51. Zhao Z, Shaw SL, Yin L, Fang Z, Yang X, Zhang F, Wu S: **The effect of temporal sampling intervals on typical human mobility indicators obtained from mobile phone location data**. *Int J Geogr Inf Sci* 2019, **33**(7):1471-1495.
52. Alarcon Falconi TM, Estrella B, Sempértegui F, Naumova EN: **Effects of Data Aggregation on Time Series Analysis of Seasonal Infections**. *Int J Env Res Pub He* 2020, **17**(16):5887.
53. Thomas MF: **Landscape sensitivity in time and space — an introduction**. *Catena* 2001, **42**(2):83-98.
54. Chen Y, Chen X, Liu Z, Li X: **Understanding the spatial organization of urban functions based on co-location patterns mining: A comparative analysis for 25 Chinese cities**. *Cities* 2020, **97**:102563.
55. Alvarez-Mendoza CI, Teodoro A, Freitas A, Fonseca J: **Spatial estimation of chronic respiratory diseases based on machine learning procedures —an approach using remote sensing data and environmental variables in quito, Ecuador**. *Appl Geogr* 2020, **123**:102273.
56. Annesi-Maesano I, Forastiere F, Balmes J, Garcia E, Harkema J, Holgate S, Kelly F, Khreis H, Hoffmann B, Maesano CN: **The clear and persistent impact of air pollution on chronic respiratory diseases: a call for interventions**. In., vol. 57: Eur Respiratory Soc; 2021.
57. Remmers T, Koolwijk P, Fassaert I, Nolles J, de Groot W, Vos SB, de Vries SI, Mombarg R, Van Kann DHH: **Investigating young children's physical activity through time and place**. *International journal of health geographics* 2024, **23**(1):12.
58. Wilhite K, Booker B, Huang B-H, Antczak D, Corbett L, Parker P, Noetel M, Rissel C, Lonsdale C, del Pozo Cruz B *et al*: **Combinations of Physical Activity, Sedentary Behavior, and Sleep Duration and Their Associations With Physical, Psychological, and Educational Outcomes in Children and Adolescents: A Systematic Review**. *American Journal of Epidemiology* 2023, **192**(4):665-679.
59. Cheng J, Yin P: **Analysis of the Complex Network of the Urban Function under the Lockdown of COVID-19: Evidence from Shenzhen in China**. *Mathematics* 2022, **10**(14):2412.
60. Davis CA, Fonseca FT: **Assessing the certainty of locations produced by an address geocoding system**. *Geoinformatica* 2007, **11**:103-129.
61. Ni Y, Shi G, Qu J: **Indoor PM2.5, tobacco smoking and chronic lung diseases: A narrative review**. *Environ Res* 2020, **181**:108910.
62. Yan M, Ge H, Zhang L, Chen X, Yang X, Liu F, Shan A, Liang F, Li X, Ma Z *et al*: **Long-term PM2.5 exposure in association with chronic respiratory diseases morbidity: A cohort study in Northern China**. *Ecotox Environ Safe* 2022, **244**:114025.
63. Gao X, Starmer J, Martin ER: **A multiple testing correction method for genetic association studies using correlated single nucleotide polymorphisms**. *Genet Epidemiol* 2008, **32**(4):361-369.
64. Tango T, Takahashi K: **A flexible spatial scan statistic with a restricted likelihood ratio for detecting disease clusters**. *Stat Med* 2012, **31**(30):4207-4218.
65. Franklin M, Vora H, Avol E, Mcconnell R, Lurmann F, Liu F, Penfold B, Berhane K, Gilliland F, Gauderman WJ: **predictors of intra-community variation in air quality hhs public access**. 2019.
66. Sandrah, P., Eckel, Kiros, Berhane, Muhammad, T., Salam, Edward, B.: **Residential Traffic-Related Pollution Exposures and Exhaled Nitric Oxide in the Children's Health Study**. *Environmental Health Perspectives* 2011, **119**(10).

Figures

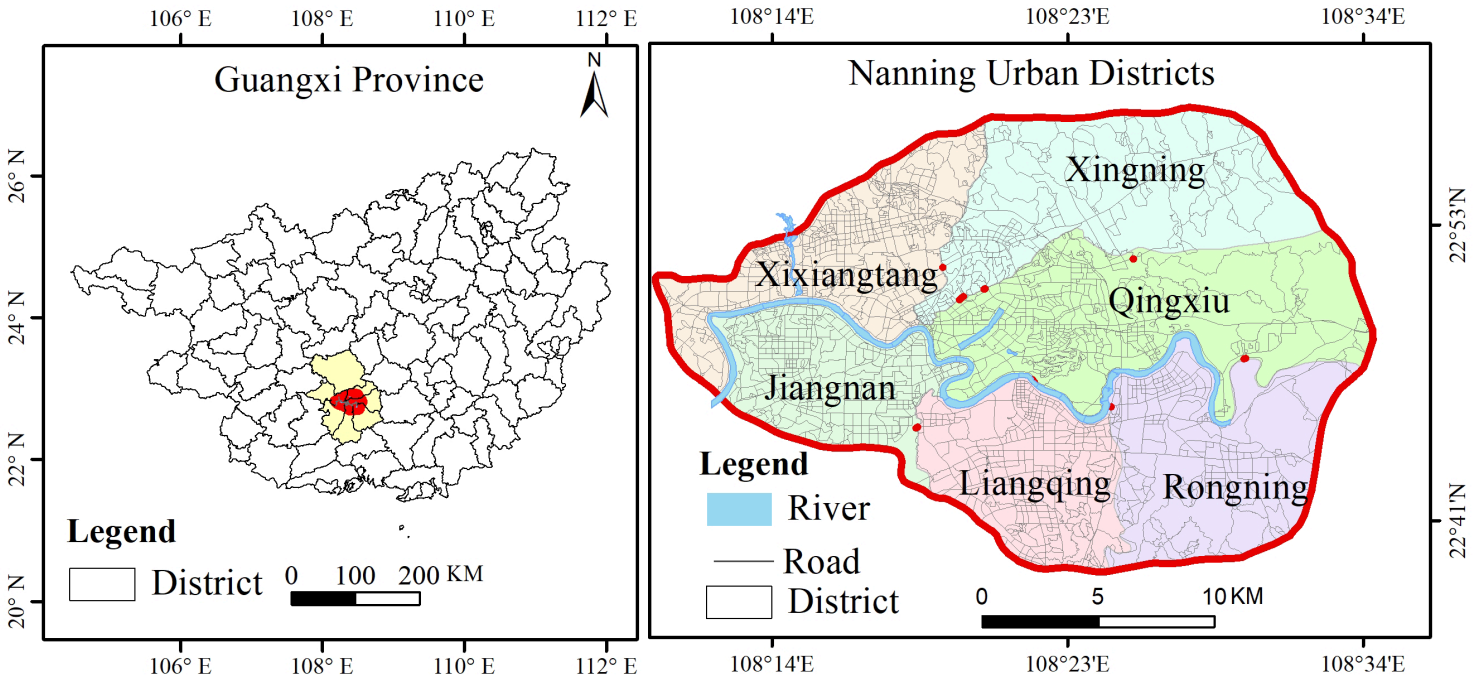


Figure 1

Location of the Nanning City study area (right) in Guangxi Province (left)

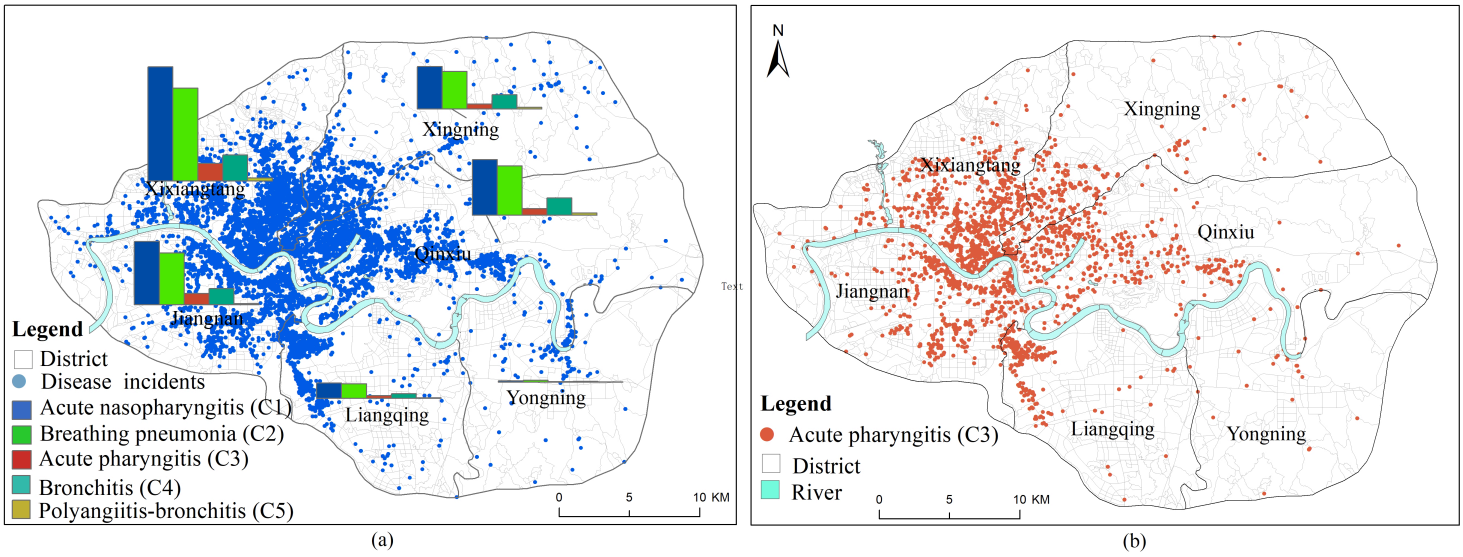


Figure 2

Distributions of (a) incidents and statistics and (b) Acute pharyngitis (C3), in the central urban area

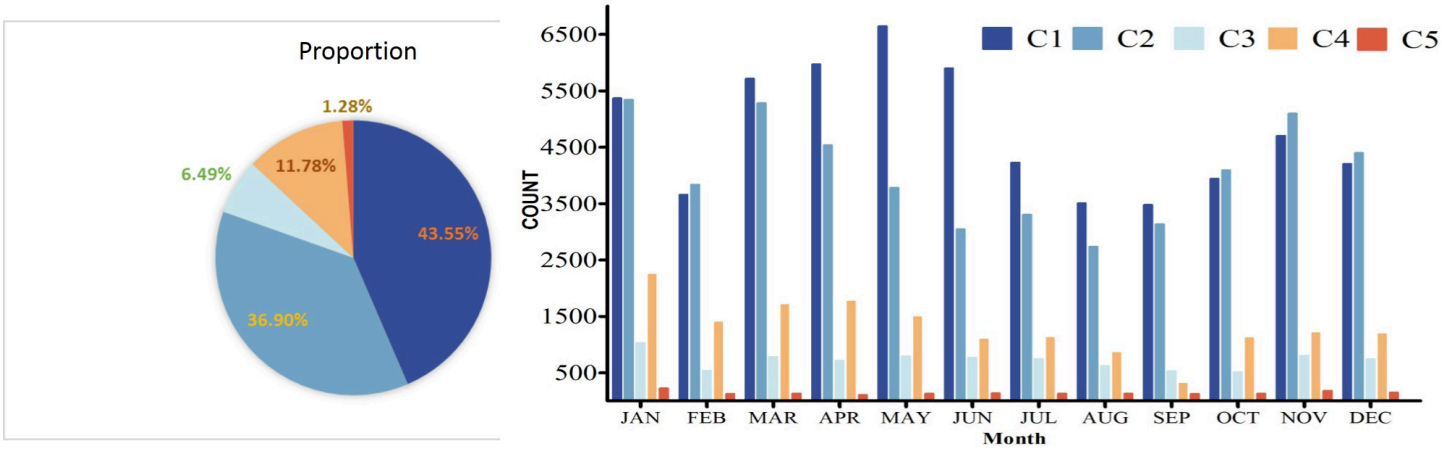


Figure 3

The statistic feature of five respiratory diseases in 2016 (total and monthly)

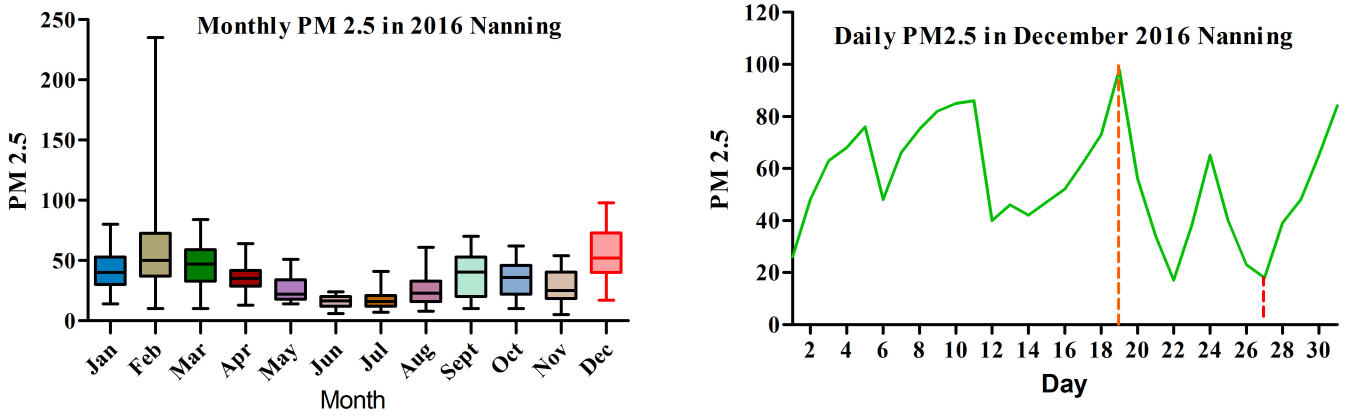


Figure 4

Monthly distribution of PM2.5 in 2016 and daily distribution in December 2016

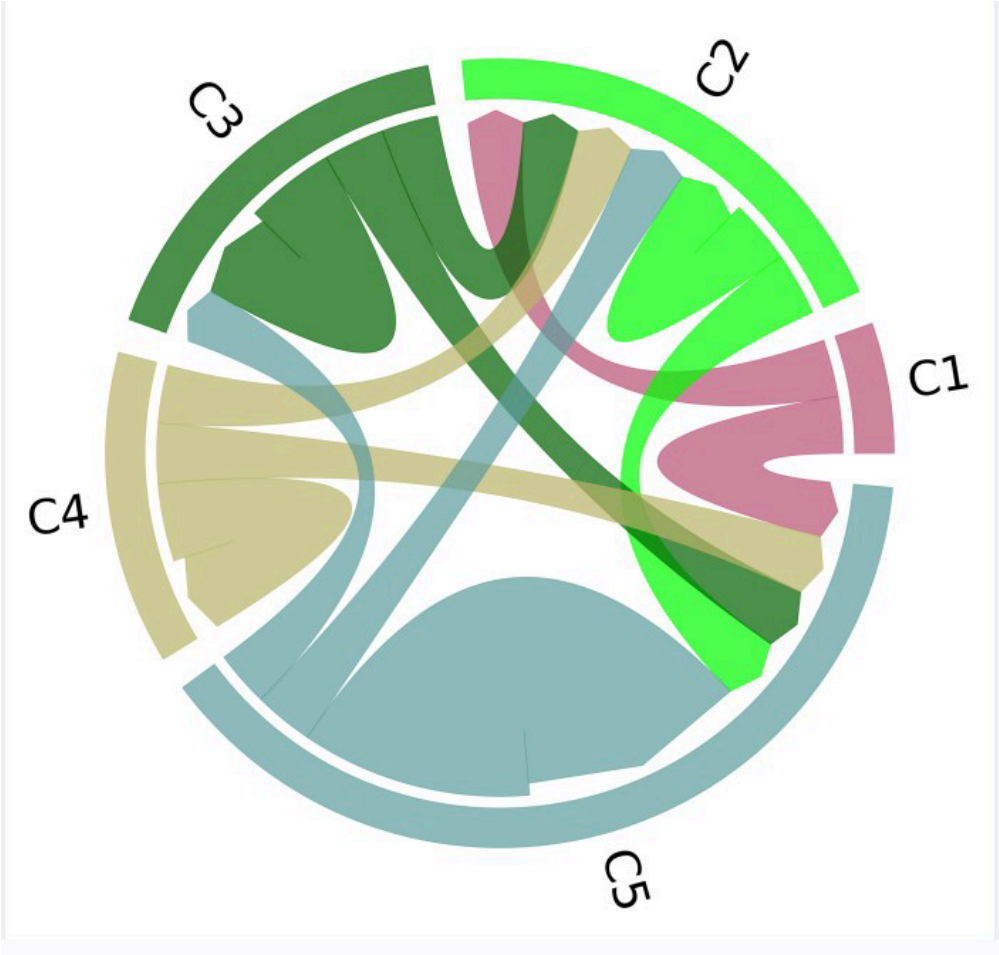


Figure 5

Perceived connections between five categories with global value large than 1.

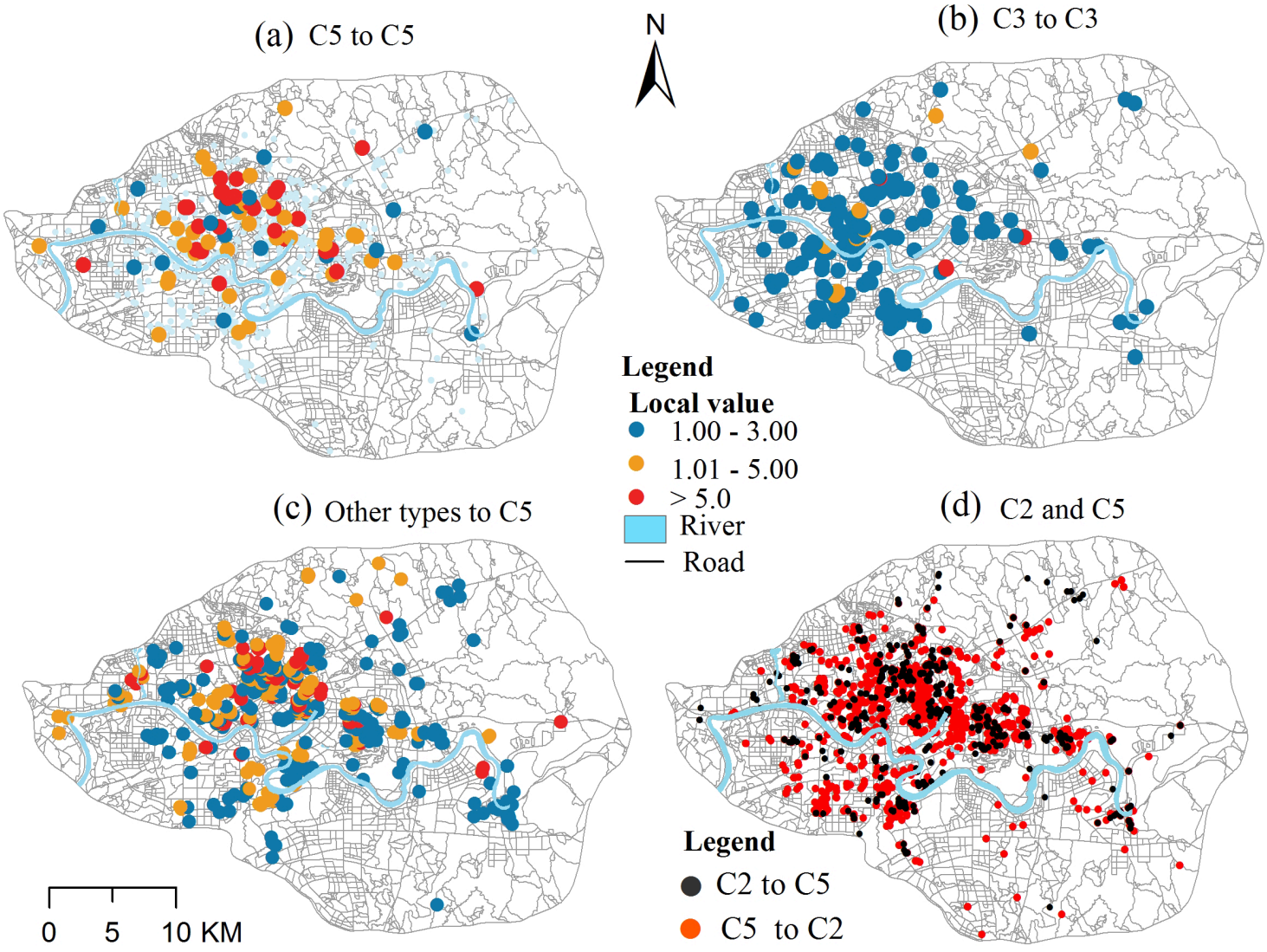


Figure 6

Local clusters from GTWCLQ12 (a) C5 to C5; (b) C3 to C3; (c) other types to C5; (d) the asymmetrical dependence between C2 and C5

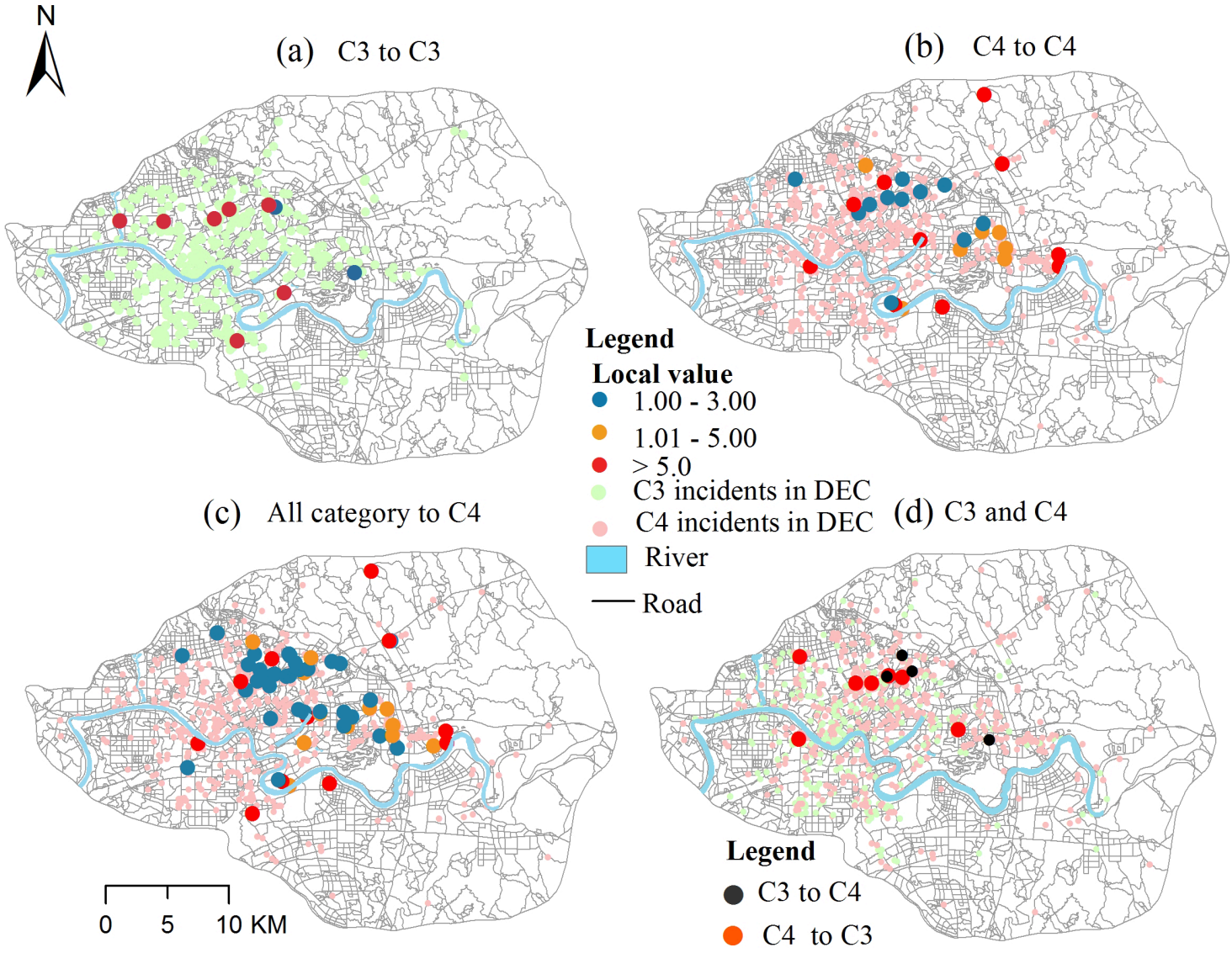


Figure 7

Local clusters from GWTCLQ31 at daily level:(a) C3 to C3; (b) C4 to C4;(c) other types to C4; (d) the asymmetrical dependence between C3 and C4

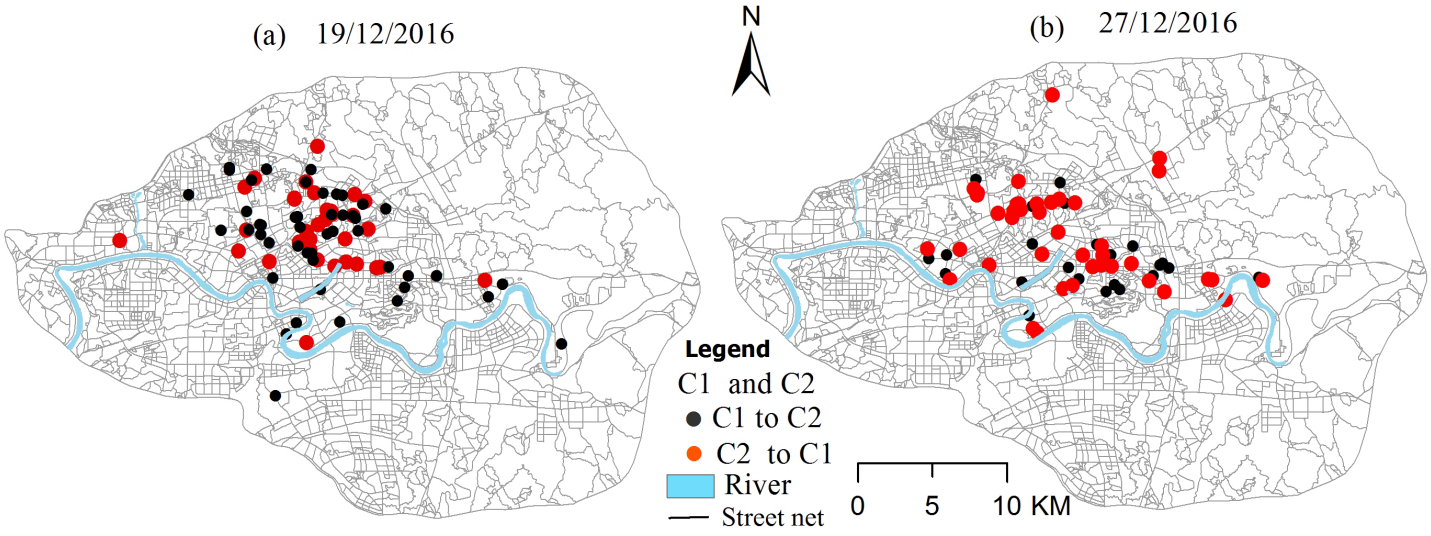


Figure 8

Symmetric co-location association between Acute nasopharyngitis (C1) and Breathing pneumonia (C2) at different air conditions